

CTU-Chat: Supervisor-Routed Multi-Agent RAG with Heterogeneous Knowledge Allocation for University Counseling

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Abstract. Student counseling in higher education draws on heterogeneous institutional knowledge, including relational curricula, cohort-specific tuition schedules, statutory exemption criteria, scholarship policies, and narrative administrative regulations. Monolithic RAG systems and loosely coordinated agent architectures can be unreliable in this setting because different forms of knowledge require different retrieval and reasoning methods. We present CTU-Chat, a centralized supervisor-routed, domain-specialized multi-agent system for Vietnamese university counseling. The supervisor directs each query to one of four specialists, while each specialist is assigned an appropriate knowledge source, retrieval path, and set of tools. Relational academic information is handled through a knowledge graph; structured tuition and scholarship information is supported by dedicated lookup and calculation functions; and narrative regulations are retrieved through hybrid document search. This division allows the system to use each source according to its structure while limiting unnecessary tool access. Evaluation shows that CTU-Chat achieves 97.0% routing accuracy and 0.8667 Hit@1 on the benchmark. These findings suggest that combining centralized coordination with domain-specific knowledge allocation can provide a practical and reliable foundation for university counseling systems.

Keywords: Multi-Agent Systems · Domain-Specialized Agents · Centralized Supervisor Routing · Heterogeneous Knowledge Allocation · Retrieval-Augmented Generation · University Counseling

1 Introduction

In large-scale universities, particularly Can Tho University (CTU)—a key higher education institution in the Mekong Delta with nearly 49,000 undergraduate students in the 2025–2026 academic year [1]—advisory needs span a wide range of

academic and administrative areas, while relevant regulations and policies can change over time. Several automated advisory systems have been developed at CTU, including CAAS, which provides text- and voice-based admissions counseling, and REBot, which answers questions about academic regulations by combining document retrieval with a knowledge graph. However, these systems focus on specific advisory areas, and their ability to support multiple domains simultaneously—including curricula, academic regulations, tuition fees, scholarships, and administrative procedures—remains limited. Because these domains have different knowledge structures, an effective system must employ appropriate representation, retrieval, and processing mechanisms for each type of information. This gap motivates the development of CTU-Chat.

University counseling presents challenges not only because it covers multiple domains, but also because institutional knowledge is represented in different forms. (1) **Academic curricula** contain hierarchical structures and prerequisite relationships; (2) **tuition schedules** are organized by admission cohort and field of study; (3) **tuition exemption policies** and (4) **scholarship policies** combine eligibility conditions with specific support rates and thresholds; while (5) **general administrative regulations** are presented in lengthy narrative documents. Therefore, no single representation or retrieval method is suitable for every domain. Relational curricula are better suited to graph-based retrieval, financial policies require structured data lookup and deterministic calculation, and narrative regulations benefit from hybrid lexical and semantic retrieval. This heterogeneity requires each domain to be matched with an appropriate knowledge representation, retrieval method, and reasoning mechanism.

To automate institutional QA, Retrieval-Augmented Generation (RAG) [8,5] and GraphRAG [3] augment large language models (LLMs) with external documents or knowledge graphs, while agentic architectures [20,18,15] introduce iterative tool invocation and multi-agent coordination. Nevertheless, applying existing paradigms to university counseling reveals significant limitations. Monolithic RAG pipelines risk a uniform representation fallacy: forcing relational curricula, tabular fee matrices, and narrative legal prose into a single retrieval index can fragment tabular data and prerequisite links. Conversely, uncoordinated multi-agent meshes can expose broad tool sets across agents, increasing the risk of tool misselection, parameter errors, and extended execution times, while known limitations in language-model arithmetic [13] motivate deterministic calculation for credit-based tuition.

These challenges crystallize three open research gaps in automated university counseling: (1) limited orchestration of domain-specialized agents with separate knowledge spaces and tool permissions; (2) the homogeneous representation of structurally different institutional knowledge; and (3) the lack of representation-aware routing to appropriate retrieval and reasoning mechanisms. These gaps are examined in greater detail in Section 2.

To address these gaps, we formulate three research questions:

- **RQ1 (Domain-Specialized Multi-Agent Architecture):** How effectively can a centralized supervisor coordinate domain-specialized agents

with isolated knowledge spaces and constrained tool access for heterogeneous university counseling tasks?

- **RQ2 (Heterogeneous Knowledge Allocation):** How does assigning different knowledge types to appropriate representations—graph-oriented access for relational academic data, structured access for tuition data, and hybrid document retrieval for narrative regulations—affect retrieval and answer quality compared with a uniform retrieval representation?
- **RQ3 (Representation-Aware Evidence Routing):** How does intent-guided routing between graph-oriented and hybrid document retrieval paths affect retrieval effectiveness, answer accuracy, and latency across different counseling domains?

We hypothesize that reliable university counseling is better supported by a supervisor-routed multi-agent architecture in which each specialist is restricted to knowledge representations, retrieval paths, and tools appropriate to its domain, rather than by a uniform retrieval-and-reasoning pipeline. Accordingly, CTU-Chat is designed around three complementary dimensions: specialist coordination, heterogeneous knowledge allocation, and representation-aware evidence routing.

To investigate these research questions, we propose CTU-Chat, a centralized supervisor-routed multi-agent architecture for heterogeneous university counseling. The system coordinates four domain-specialized agents for **academic, financial, scholarship, and general administrative inquiries**. Each specialist is assigned an isolated knowledge space, an appropriate retrieval path, and constrained tool privileges according to the structure of its domain knowledge. This design enables CTU-Chat to combine graph-based retrieval, structured data access, deterministic calculation, and hybrid document retrieval within a unified counseling framework.

For clarity, our contributions are organized into three corresponding components, denoted as **C1**, **C2**, and **C3**, which respectively address RQ1, RQ2, and RQ3. These contributions are described in detail in Section 3 and evaluated through the corresponding experimental scenarios in Section 4.

The remainder of this paper is organized as follows. **Section 2** reviews related work on retrieval methods, knowledge graphs, agentic RAG, and educational question-answering systems and summarizes the research gaps. **Section 3** presents the proposed CTU-Chat architecture and its domain-specialized agents. **Section 4** describes the retrieval, evidence-ranking, and deterministic calculation methods. **Section 5** reports the experimental setup, evaluation results, and discussion. Finally, **Section 6** concludes the paper and outlines directions for future work.

2 Related Work

This section reviews established retrieval and reasoning paradigms in the literature, examining their operational mechanisms and limitations in institutional counseling.

2.1 Vietnamese NLP Foundations and Linguistic Specifics

Unlike whitespace-delimited languages, Vietnamese is an isolating, tonal language where white spaces serve as syllable separators rather than strict word boundaries [10]. Over 85% of Vietnamese lexical units are multi-syllabic compound words (e.g., “*học phí*” [tuition fee]), which introduces severe morphological ambiguity under standard whitespace tokenization and motivates dedicated syllable segmentation [16].

To capture semantic representations beyond lexical surface forms, domain-adapted language models such as PhoBERT [10] and pretrained bi-encoders project Vietnamese text into dense semantic spaces. However, Vietnamese higher education regulations rely heavily on rigid Sino-Vietnamese administrative terminology alongside explicit cohort and decree codes. This dual characteristic motivates coupling syllable-aware dense embeddings with exact lexical token matching to avoid both semantic drift and alphanumeric recall failures.

2.2 Sparse and Dense Retrieval

Lexical Retrieval (BM25): Sparse keyword search using BM25 [14] evaluates term frequency (tf) and inverse document frequency (IDF) over an inverted index:

$$s_{\text{BM25}}(q, d) = \sum_{w \in q} \text{IDF}(w) \cdot \frac{\text{tf}(w, d) \cdot (k_1 + 1)}{\text{tf}(w, d) + k_1 \cdot \left(1 - b + b \cdot \frac{|d|}{|d|}\right)}, \quad (1)$$

with $k_1 = 1.5$ and $b = 0.75$. BM25 supports exact lexical matching for rigid alphanumeric tokens (decision codes, cohort labels, course identifiers), but may miss semantic paraphrases.

Dense Semantic Retrieval: Dense retrieval systems [7,8] map queries and passages into embedding spaces using bi-encoders, computing relevance via vector similarity. While dense models capture paraphrastic intent, rare alphanumeric identifiers and out-of-vocabulary regulatory tokens can remain difficult and may introduce cross-cohort noise, motivating hybrid retrieval.

2.3 Hybrid Retrieval and Reciprocal Rank Fusion

To merge BM25 and dense candidates without score calibration, Reciprocal Rank Fusion (RRF) [2] sorts documents by reciprocal position: $s_{\text{RRF}}(d) = \sum_{s \in \mathcal{M}} \frac{1}{k + r_s(d)}$ ($k = 60$), while neural cross-encoders [11,19] jointly encode query–passage pairs for reranking. Adaptive-RAG further demonstrates that retrieval strategies can be selected according to question complexity [6]. These approaches motivate dynamic retrieval, but do not directly allocate institution-specific relational, tabular, and narrative knowledge to isolated specialist paths.

2.4 Knowledge Graphs and GraphRAG

Knowledge graphs represent domain knowledge as structured triples (h, r, t) [12]. GraphRAG [3] leverages graph topology to uncover multi-hop associative paths escaping flat chunking, benefiting curriculum structures and prerequisite chains. However, universal graph extraction across institutional corpora produces hallucinated relationships, while forcing narrative legal prose into graphs is structurally unnatural, motivating selective rather than universal graph use.

2.5 Agentic RAG and Multi-Agent Orchestration

ReAct demonstrates interleaved reasoning and action, AutoGen provides a framework for multi-agent conversation, and Toolformer studies when and how language models invoke external APIs [20,18,15]. For institutional counseling, these capabilities raise design risks when applied without explicit constraints: broad tool exposure can increase tool and parameter errors; peer-to-peer exchanges can increase coordination cost; a shared index can mix evidence across domains; and a common retrieval path can obscure opportunities for direct graph traversal or structured lookup. These risks motivate the bounded, supervisor-routed specialization evaluated in this work.

2.6 Educational and University Counseling Systems

Educational chatbots primarily assist in informational, advising, and learning-support roles, though evaluation and adaptation remain challenging [17]. Recent institutional counseling systems like REBot [9] combine document retrieval with category-guided GraphRAG for academic regulations. However, CTU-Chat departs from prior advisory frameworks across three core dimensions:

- **Task Scope and Deterministic Arithmetic:** Unlike prior systems restricted to single-turn narrative text-QA over general regulations, CTU-Chat models 113 undergraduate curricula as relational property graphs and incorporates external deterministic arithmetic tools (`tin_h_toan_hoc_phi`, `tin_h_tien_hoc_bong`) for conditional tuition schedules and statutory exemptions.
- **Multi-Agent Routing vs. Monolithic Pipelines:** Rather than using monolithic pipelines driven by shallow intent classifiers, CTU-Chat deploys a LangGraph `StateGraph` supervisor that routes queries to four bounded specialists with isolated evidence spaces, dedicated retrieval paths (hybrid RRF vs. Cypher graph traversal), and asymmetric tool privileges.
- **Metric Incommensurability:** While prior work measures classification accuracy and F_1 -scores, CTU-Chat operates as an end-to-end multi-agent generative system with programmatic tool invocation. Consequently, direct metric comparison is inapplicable; we instead evaluate generation faithfulness using Ragas alongside bounded tool execution benchmarks.

2.7 Comparative Synthesis and Research Gaps

Synthesizing established paradigms highlights clear trade-offs across institutional counseling workflows [2,3,6,7,14,15,18,20]: lexical matching ensures exact identifier recall but misses paraphrases; dense embeddings capture semantics but cause cross-cohort confusion; GraphRAG models relations but degrades on unstructured text; and unconstrained multi-agent frameworks lack execution determinism. These trade-offs crystallize three core research gaps:

- **Gap 1 (Lack of Bounded Domain-Specialized Agent Orchestration):** Prior frameworks rely on monolithic pipelines or unconstrained agent meshes without execution boundaries, increasing the risks of tool mis-selection, parameter hallucination, and high coordination costs.
- **Gap 2 (The Uniform Representation Fallacy):** Forcing relational curricula, rate matrices, and regulatory prose into a uniform retrieval index fragments prerequisite chains and induces relation hallucinations across structurally incompatible evidence spaces.
- **Gap 3 (Absence of Representation-Aware Routing and Deterministic Computation):** Existing systems cannot conditionally steer inquiries to dedicated representation paths (e.g., direct Cypher traversal vs. hybrid document RAG) nor decouple arithmetic statutory calculations from generative hallucination.

These limitations motivate our centralized supervisor-routed architecture and heterogeneous knowledge allocation detailed in Section 3.

3 Proposed Model

This section presents the architecture, algorithms, and evidence-acquisition mechanisms of CTU-CHAT. The proposed model is organized around three design principles: *centralized agent coordination*, in which a supervisor dispatches each query to one domain agent without peer-to-peer delegation; *heterogeneous knowledge allocation*, which assigns each information structure to a corresponding evidence representation; and *representation-aware evidence routing*, which selects either a direct graph-tool path or an intent-scoped retrieval path according to the assigned domain. Here, each *specialist* represents an autonomous domain agent configured with dedicated prompt constraints, an assigned evidence store, and a restricted tool registry.

3.1 Overall Proposed Architecture

Figure 1 illustrates the system architecture of CTU-CHAT, structured across three coordinated tiers: (1) the *Interaction and Centralized Supervisor Tier*, (2) the *Domain-Specialized Multi-Agent Core Layer*, and (3) the *Heterogeneous Knowledge Storage and Tool Tier*.

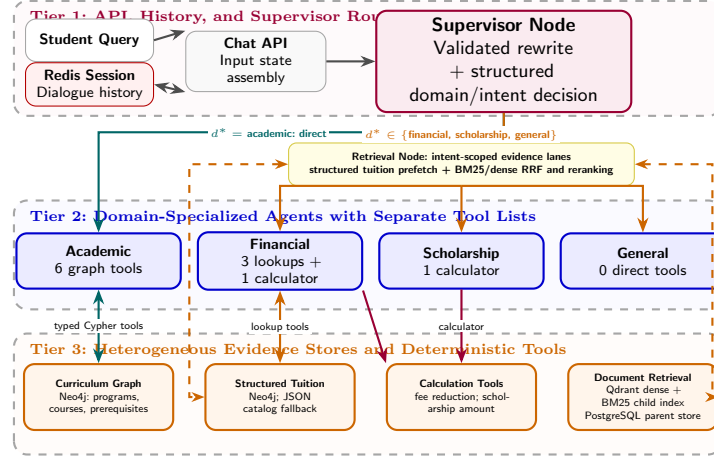


Fig. 1. Implemented CTU-CHAT architecture. The Chat API loads dialogue history from Redis and supplies it in the graph input state. The first conditional edge sends academic queries directly to the graph-enabled agent and sends the other domains through intent-scoped retrieval; the second edge dispatches the retrieved evidence to the selected domain-specialized agent. Solid agent-resource links denote direct tool access, whereas dashed links denote evidence acquisition performed by the retrieval node.

Application-Level Specialization of LangGraph StateGraph: Rather than modifying the underlying LangGraph engine, CTU-CHAT establishes an application-level orchestration policy governed by a typed state contract, isolated execution paths, and partitioned tool registries. Following supervisor dispatch, the four domain specialists operate independently without peer-to-peer delegation, selecting actions strictly from their construction-time tool bindings.

- **Node-scoped state updates:** `AgentState` defines explicit write boundaries: the supervisor writes to the routing fields (`search_query`, `next_agent`, and `routing_decision`); the retrieval node updates `context`; and each specialist returns `response`. This contract guarantees transparent data ownership and eliminates cross-domain context pollution.
- **Two-Tier Conditional Routing Edges:** Instead of letting agents dynamically negotiate execution flow, the orchestration topology is governed by deterministic conditional functions:

$$\text{Edge}_{\text{sup}}(\mathcal{S}) = \begin{cases} \text{'academic_agent'}, & \text{if } \mathcal{S}[\text{'next_agent'}] = \text{'academic'} \\ \text{'retrieval'}, & \text{otherwise,} \end{cases} \quad (2)$$

bypassing document retrieval for queries assigned to the academic agent. At the retrieval exit, $\text{Edge}_{\text{ret}}(\mathcal{S})$ dispatches the retrieved context to the selected financial, scholarship, or general agent.

- **Acyclic outer orchestration:** The StateGraph has no peer-to-peer agent edges. Each selected domain-specialized agent returns to `END`, so the outer

routing path cannot cycle. ReAct tool interactions inside an agent remain distinct from this outer graph topology.

- **Domain-specialized agents with asymmetric tools:** The academic, financial, and scholarship agents are constructed with separate prompts and tool lists, while the general agent has no direct tool. This confines tool selection to the agent selected by the supervisor and keeps domain-specific evidence paths separate.

3.2 Domain-Specialized Agents and Asymmetric Privilege Allocation

Table 1 details the tool bindings and evidence access paths assigned to each specialist. CTU-CHAT decouples macro-level domain routing from micro-level action selection: the centralized supervisor dispatches the query to a specific domain, while the assigned specialist operates strictly within its preconfigured tool boundary.

Table 1. Operational profiles and direct tool availability of the domain-specialized agents.

Agent role	Evidence store	Access path
Academic	Neo4j	ReAct over graph tools
Financial	Neo4j, JSON, document RAG	Retrieval plus calculation
Scholarship	Document RAG	Retrieval plus calculation
General	Document RAG	Evidence-grounded synthesis

Direct tools vs. pre-retrieved context: The academic agent directly selects among six graph tools. The financial agent receives intent-scoped context and may select among three structured lookup tools and `tinhtuan.hoc_phi`; the scholarship agent receives retrieved policy context and has only `tinhtien.hoc_bong`; and the general agent performs evidence-grounded synthesis without direct tools. This asymmetric tool assignment enforces architectural guardrails across agents, restricting tool invocation to valid domains and preventing uncoordinated tool proliferation.

3.3 Neo4j Knowledge Graph Specification and Cypher Tool Mechanics

Schema defined by the ingestion layer: The Neo4j graph comprises 10 node labels (7 for curriculum: `Program`, `Course`, `Block`, etc.; and 3 for tuition: `TuitionFee`, `TuitionPolicy`, `ExemptionBasisRate`) connected via 13 relationship types (`HAS_BLOCK`, `REQUIRES`, etc.). Curriculum structure spans seven labels, while tuition and exemption data span three. Narrative regulations remain in the document-retrieval store unless represented by one of these structured entities.

The graph schema models the essential relational dependencies required for academic counseling, capturing program composition, prerequisite chains, learn-

ing outcomes, and program-to-tuition links without introducing unnecessary ontological overhead.

The ingestion layer defines thirteen relationship types: `HAS_BLOCK`, `CONTAINS`, `REQUIRES`, `PARALLEL_WITH`, `HAS_PLO`, `HAS_PEO`, `REALIZED_BY`, `HAS_TRACK`, `BELONGS_TO_TRACK`, `HAS_JOB_POSITION`, `HAS_TUITION`, `GOVERNED_BY`, and `HAS_EXEMPTION_BASIS`. They encode program composition, course dependencies, learning outcomes, specialization and career links, and financial-policy associations.

Cypher tool mechanics: The system does not ask the LLM to generate arbitrary Cypher. After the supervisor selects the academic agent, its ReAct policy chooses one of six typed tools and supplies the tool arguments. The tool normalizes identifiers or names and calls a fixed, parameterized Cypher query in the graph service. The tools implement three query classes:

- *Attribute lookup and search:* `tra_cuu_nganh` and `tim_nganh` match program codes or normalized program, unit, and course names; `so_sanh_nganh` applies the same lookup before comparing two programs.
- *Structural traversal:* `xem_chuoi_tien_quyet` follows `REQUIRES` paths up to ten hops and returns direct `PARALLEL_WITH` courses. `mon_chung_giua_nganh` computes a course intersection across two program–block paths, and `tim_nganh_co_mon` traverses those paths in reverse.
- *Structured financial lookup:* parameterized queries match `TuitionFee`, `TuitionPolicy`, and `ExemptionBasisRate` properties and their program relationships. This graph lookup is separate from the BM25–dense hybrid document retrieval path used for narrative regulations.

This design binds agent actions directly to deterministic graph operations: the LLM only selects the appropriate tool and supplies its arguments, while the query topology remains fixed and parameterized within the graph service. As a result, the multi-agent workflow reliably incorporates graph evidence without the fragility of unconstrained Cypher generation.

3.4 Algorithmic Foundations: Routing and Deterministic Arithmetic

Algorithm 1 formalizes the operational core of CTU-CHAT: supervisor-guided evidence routing and deterministic financial calculation outside the generative LLM. It details the routing and dispatch pipeline. Contextual rewriting is validated against recent dialogue history to prevent semantic drift, while structured classification resolves the target domain and intent, with an explicit fallback to the general agent when classification is ambiguous.

Algorithm 1: Supervisor Routing and Conditional Evidence-Path Dispatch

Input: Student query q , Dialogue history $\mathcal{H}$, Shared state $\mathcal{S}$
Output: State updates and next graph node

```

1  $q_{\text{standalone}} \leftarrow q$ ;
2  $q_{\text{prev}} \leftarrow \text{MostRecentUserQuery}(\mathcal{H})$ ;
3 if ShouldRewriteQuery( $q, q_{\text{prev}}$ ) then
4    $q_{\text{cand}} \leftarrow \text{FastLLM\_Rewrite}(q, q_{\text{prev}})$ ;
5   if ValidateRewrite( $q, q_{\text{cand}}, q_{\text{prev}}$ ) then
6      $q_{\text{standalone}} \leftarrow q_{\text{cand}}$ ;
7  $\langle d^*, \iota^* \rangle \leftarrow \text{StructuredSupervisor}(q)$ ;
8 if structured classification fails then
9    $d^* \leftarrow \text{'general'}$ ;  $\iota^* \leftarrow \text{'other'}$ ;
10 Update  $\mathcal{S}[\text{search\_query}] \leftarrow q_{\text{standalone}}$ ,  $\mathcal{S}[\text{next\_agent}] \leftarrow d^*$ , and
     $\mathcal{S}[\text{routing\_decision}] \leftarrow \text{QueryRoutingDecision}(\iota^*)$ ;
11 if  $d^* = \text{'academic'}$  then
12    $v_{\text{next}} \leftarrow \text{'academic\_agent'}$ ;
13 else
14   Build retrieval lanes from  $\iota^*$  and retrieve evidence;
15    $v_{\text{next}} \leftarrow d^*_{\text{-agent}}$ ;
16 return  $\langle \mathcal{S}, v_{\text{next}} \rangle$ ;

```

Routing characteristics: Algorithm 1 separates three decisions that a monolithic pipeline would conflate: whether conversational rewriting is needed, which domain agent should answer, and which evidence path should run. Rewrite validation limits unsupported additions to a context-dependent query; structured output constrains routing to the declared domains and intents; and the explicit **general/other** fallback keeps classification failure on a defined path. Conditional edges then make the selected path inspectable without allowing peer-to-peer agent delegation.

Deterministic Tuition Reduction Arithmetic: For fee-reduction requests, CTU-CHAT augments the LLM with the external deterministic tool `tinh_toan_hoc_phi`. The LLM interprets the request and supplies three parameters—the actual per-credit tuition T , the per-credit exemption basis B , and the percentage entitlement p —but does not perform the arithmetic through free-form text generation. The external tool validates the percentage range and applies the fixed calculation

$$D = B \frac{p}{100}, \quad (3)$$

$$P = \max(0, T - D), \quad (4)$$

where D is the reduction per credit and P is the remaining amount per credit. Monetary inputs are expected in VND per credit. The tool enforces $0 \leq p \leq 100$ and clamps P at zero; values outside the percentage range return an error. Returned monetary values are formatted as integers. Delegating this arithmetic to an external deterministic tool ensures reproducible calculations, prevents

floating-point inaccuracies, and eliminates arithmetic hallucination from the generative model.

Advantages of the Algorithmic Separation:

- *External tool augmentation:* The LLM handles natural-language interpretation, while a deterministic component performs validation and arithmetic. This extends the system’s calculation capability without relying on unconstrained numerical generation by the LLM.
- *Distinct monetary bases:* The calculation keeps actual tuition T separate from exemption basis B , applies the percentage only to B , validates its range, and prevents a negative payable amount through explicit clamping.

3.5 Heterogeneous Knowledge Allocation and Hybrid Retrieval

This layer implements *heterogeneous knowledge allocation*: rather than forcing all institutional knowledge into a uniform document index, CTU-CHAT assigns each domain agent a representation and access path suited to the structure of its evidence. Relational curriculum knowledge is preserved in a graph for property lookup and traversal, cohort-dependent tuition data is accessed through structured lookup, and narrative regulations and scholarship policies remain in document retrieval. The resulting allocation is intentionally asymmetric across agents, embedding representation choices directly into the executable routing policy.

Table 2 makes this policy explicit by separating the physical evidence representation from the mechanism used to access it at runtime.

Table 2. Decision criteria mapping query characteristics to evidence representation mechanisms.

Query characteristic	Evidence path	Rationale
Program/course attributes	Neo4j property lookup	Exact or normalized match
Prerequisites/shared courses	Neo4j traversal	Stored graph relationships
Regulation identifiers	BM25 child index	Exact token match
Paraphrased regulations	Hybrid RRF + reranker	Lexical + semantic signals
Fee/scholarship calculation	Python tool	Fixed formula outside LLM

At runtime, this allocation is enforced by the StateGraph rather than left to unrestricted agent choice. Academic queries bypass the document-retrieval node and enter the graph-enabled academic agent directly. Financial, scholarship, and general queries enter the retrieval node, where the supervisor-assigned intent determines the metadata-filtered retrieval lanes before the resulting context is passed to only the selected domain agent. Thus, knowledge allocation and evidence-path selection operate as separate but tightly coordinated design decisions.

Hybrid Document Retrieval Path. For narrative regulations and scholarship criteria, evidence is acquired through the following pipeline:

1. *Child-level candidate generation.* A BM25 index searches child chunks, while Qdrant performs dense retrieval over their 768-dimensional embeddings.
2. *Parent resolution.* Parent identifiers associated with the retrieved child chunks are used to load the corresponding parent sections from PostgreSQL.
3. *Reciprocal Rank Fusion.* The dense- and BM25-derived parent rankings are combined as

$$s_{\text{RRF}}(d) = \sum_{s \in \{\text{bm25}, \text{dense}\}} \frac{1}{k + r_s(d)}, \quad k = 60.$$

4. *Cross-encoder reranking.* The fused parent candidates are reranked using BAAI/bge-reranker-v2-m3, and up to six parent sections are supplied to the selected domain agent.

Parent sections use a 2,800-character target with 100-character overlap; child chunks use a 400-character target with 50-character overlap.

Architectural Synthesis: In summary, CTU-CHAT integrates established retrieval components (BM25, dense embeddings, RRF, and neural cross-reranking) exclusively within the narrative regulation path. By coordinating these document pipelines with graph traversals and deterministic arithmetic engines under a centralized supervisor, the architecture ensures that each administrative inquiry is resolved through its structurally optimal evidence modality.

4 Experimental Results

This section presents the empirical evaluation of CTU-CHAT, detailing the benchmark suites, infrastructure, evaluation metrics, three experimental scenarios, discussion answering each research question, and threats to validity.

4.1 Experimental Benchmark and Protocols

To ensure rigorous validation and avoid data contamination, our empirical evaluation is established upon a corpus of 200 curated administrative queries grounded in official Can Tho University (CTU) statutes. Following standard benchmark protocol, this corpus is partitioned into: (1) **Development Benchmark** ($N = 150$ queries): Spanning 9 institutional categories (`actual_tuition`, `academic_rules`, `scholarship`, `student_loan`, `other`, `social_support`, `academic_program`, `exemption_policy`, `exemption_basis`), used for offline index calibration, prompt engineering, and qualitative failure analysis; and (2) **Held-Out Test Set** ($N = 50$ queries): A strictly separated, out-of-distribution evaluation suite covering 6 operational lanes (40% tuition, 20% academic regulations, 10% scholarships, 10% student credit loans, 10%

tuition exemptions, and 10% administrative procedures). All reported primary benchmark metrics in Tables 4 and 5 are evaluated strictly on this blind Held-Out Test Set.

Execution and Evaluation Invariants: Generative and automated evaluation tasks were executed using Google Gemini 2.5 Flash Lite (`gemini-2.5-flash-lite`) via Google Cloud Vertex AI under strict zero-temperature decoding (temperature = 0.0) to guarantee determinism. To eliminate stochastic variance, each query was evaluated across $R = 3$ independent repetitions, yielding 1,050 end-to-end generation and Ragas evaluation turns (7 configurations $\times$ 50 queries $\times$ 3 reps).

Computational Infrastructure: Neural cross-encoder reranking was executed on an NVIDIA GeForce RTX 4090 GPU (24 GB VRAM). Relational curricula are hosted in Neo4j 5 Enterprise with APOC extensions; document parent chunks (2,800 characters, 100 overlap) reside in PostgreSQL, while child chunks (400 characters, 50 overlap) are indexed in Qdrant with `bkai-foundation-models/vietnamese-bi-encoder` (768-d). Neural candidate reranking uses BAAI/`bge-reranker-v2-m3`. Multi-agent coordination was orchestrated via LangGraph under a StateGraph supervisor with Redis 7 caching.

4.2 Evaluation Metrics

Multi-Agent Coordination Metrics (Scenario 3): Following isolated evaluation protocols, we assess: (1) **Agent Accuracy & Macro-F1:** proportion of queries routed to the target specialist alongside macro-averaged F1 across the four domains; (2) **Intent Accuracy:** accuracy of fine-grained intent classification; (3) **Tool Selection, Argument Exact Match (EM), and Result Accuracy:** whether the correct execution path is invoked, whether extracted arguments match ground truth with exact numeric and string equality, and whether outputs are mathematically exact; (4) **End-to-End Pass Rate:** joint success across tool selection, argument binding, and execution output; (5) **Robustness & Tool-Gating Accuracy:** correct tool triggering on complete inputs versus deliberate tool suppression on malformed or policy-only queries; (6) **Safe Result Behavior:** explicit abstention or clarification without parameter fabrication; and (7) **Bounded Completion Rate:** proportion of decisions terminating within operational bounds (`max_tool_calls = 1`, `timeout = 60.0 s`).

Retrieval Quality Metrics (Scenario 1): Evaluated over top- k returned passages against gold document identifiers: (1) **Hit@ k** ($k \in \{1, 3\}$): proportion of queries where at least one gold document is retrieved within top- k ; (2) **Precision@5 (P@5)** and **Recall@5 (R@5)**: proportion of relevant passages in top-5, and proportion of gold passages retrieved; and (3) **Mean Reciprocal Rank (MRR@10)**: reciprocal rank of the first relevant document.

End-to-End Generation Metrics (Scenario 2): Evaluated using Ragas [4] and ground-truth alignment: (1) **Answer Relevancy (AR)**: whether the generated answer directly addresses the question; (2) **Context Recall (CR)**: proportion of factual information required by the reference answer present in the retrieved Top-7 context; (3) **Context Precision (CP)**: concentration of

relevant passages at higher ranks in Top-7; (4) **Answer Correctness (AC)**: factual agreement between generated answer and ground truth; (5) **Factual Exact Match (Fact EM)**: strict numerical and cohort exact matching on unseen queries; and (6) **Source AP**: average precision of cited regulatory sources.

4.3 Experimental Scenarios and Benchmark Results

Table 3 maps each research gap and question to its corresponding architectural component, experiment, and empirical evidence.

Table 3. Systematic mapping between Research Gaps, RQs, Proposed Modules, and Target Metrics.

Gap & RQ	Proposed Architectural Component	Experiment	Primary Metrics	Key Empirical Evidence
RQ1 (Multi-Agent)	Supervisor + 4 Domain Specialists (C1)	Scenario 3	Routing Acc., Tool EM, Safety	97.0% Routing Acc., 100% Bounded
RQ2 (Allocation)	Neo4j Graph vs. Hybrid RAG (C2)	Scenario 2	Context Precision, Ans. Correctness	T4 top Fact EM (62.03%), CP (0.6977)
RQ3 (Routing)	Subspace Governance + Reranking (C3)	Scenario 1	Hit@k, Recall@5, MRR@10	E5 attains 0.9600 Hit@1, 0.9700 MRR@10

Scenario 1 — Retrieval Component Stacking (E1–E5): *Objective & RQ Alignment:* Scenario 1 investigates **RQ3** and supports **RQ2** by evaluating the empirical contribution of progressively stacking retrieval components over the 50-query Held-Out Test Set. *Configurations:* We compare five cumulative setups: (1) **E1 (BM25 Lexical Baseline)**: BM25 search over tokenized child chunks; (2) **E2 (Dense Vector Semantic)**: 768-d vietnamese-bi-encoder vector search in Qdrant; (3) **E3 (Hybrid RRF)**: BM25 and Dense fused via Reciprocal Rank Fusion ($k = 60$); (4) **E4 (Hybrid + Neural Reranker)**: E3 augmented with `bge-reranker-v2-m3` cross-attention; and (5) **E5 (CTU-Chat Proposed Retrieval)**: integrating IG-SRR subspace routing, Neo4j relational grounding, and parent-child chunk mapping.

Results & Interpretation: As shown in Table 4, BM25 achieves a strong initial lexical baseline (0.9000 Hit@1 and 0.9333 MRR@10) due to exact cohort codes and decree identifiers. Dense retrieval alone exhibits marked degradation (0.3800 Hit@1, 0.5237 MRR@10) because vector embeddings conflate regulations across cohorts. Unreranked hybrid fusion (E3) balances lexical and semantic representations (0.6800 Hit@1, 0.7912 MRR@10). Introducing cross-encoder reranking (E4) substantially improves candidate reordering (0.8800 Hit@1, 0.9267 MRR@10, and 0.9800 Hit@3). Finally, full heterogeneous integration with IG-SRR subspace routing and Neo4j relational grounding (E5) achieves the highest performance across all primary metrics: **0.9600 Hit@1** (48 out of 50 queries retrieving gold passages at rank 1), **0.9800 Hit@3**, and **0.9700 MRR@10**, outperforming BM25 by +6.0% in top-1 accuracy and verifying robust generalization on unseen queries.

Scenario 2 — End-to-End QA Generation and Modular Ablation (T1–T7): *Objective & RQ Alignment:* Scenario 2 addresses **RQ2** and **RQ3** by evaluating end-to-end answer synthesis and modular ablation across seven configurations on the Held-Out Test Set (1,050 total evaluation turns across 3 independent repetitions). *Configurations:* (T1) BM25+LLM; (T2) Dense+LLM;

Table 4. Scenario 1: Progressive retrieval component stacking on the Held-Out Test Set ($N = 50$).

Retrieval Configuration	Hit@1	Hit@3	P@5	Recall@5	MRR@10
E1: BM25 Lexical Baseline	0.9000	0.9400	0.2320	0.9667	0.9333
E2: Dense Vector Semantic	0.3800	0.6600	0.1560	0.6433	0.5237
E3: Hybrid RRF (BM25 + Dense)	0.6800	0.9200	0.2160	0.9133	0.7912
E4: Hybrid + Neural Reranker	0.8800	0.9800	0.2200	0.9433	0.9267
E5: CTU-Chat Proposed Retrieval	0.9600	0.9800	0.2200	0.9433	0.9700

Table 5. Scenario 2: End-to-end QA generation and modular ablation on Held-Out Test Set ($N = 50$, $R = 3$, 1,050 total evaluation turns, Top-7 context).

Cfg.	Ablation Variant Description	Fact EM	CP	CR	AR	AC	Source AP
T1	BM25 Lexical Baseline	0.5572	0.6674	0.8800	0.6151	0.6735	0.8954
T2	Dense Semantic Baseline	0.5007	0.2770	0.3400	0.2809	0.3841	0.4651
T3	Hybrid RRF Baseline	0.5451	0.4339	0.7733	0.5754	0.6016	0.7490
T4	CTU-Chat Full Stack (Proposed)	0.6203	0.6977	0.8933	0.6195	0.6637	0.9311
T5	w/o Cross Reranker (bge-m3)	0.5823	0.6297	0.8133	0.5263	0.6136	0.8168
T6	w/o Neo4j Knowledge Graph Grounding	0.5711	0.5683	0.8600	0.6173	0.6582	0.9011
T7	w/o Subspace Governance (Ungoverned)	0.6089	0.6947	0.8600	0.5815	0.6430	0.9311

Metrics: Fact EM: Factual Exact Match on tuition/credits; CP: Context Precision; CR: Context Recall; AR: Answer Relevancy; AC: Answer Correctness; Source AP: Source Average Precision. Normalized to $[0, 1]$; higher is better.

(T3) Hybrid+LLM; (T4) Proposed CTU-CHAT Full Stack; (T5) w/o Reranker; (T6) w/o Neo4j Knowledge Graph; and (T7) w/o Governance Filter (removing subspace routing). *Results & Interpretation:* As reported in Table 5, the proposed full system (T4) decisively outperforms all baselines in factual fidelity: T4 attains the top **Factual Exact Match (Fact EM: 62.03% vs. 55.72% for BM25, a +6.31% absolute gain)**, demonstrating superior precision on complex numeric tuition schedules and credit rules. In addition, T4 leads in Context Precision (0.6977 vs. 0.6674 for BM25), Context Recall (0.8933 vs. 0.8800), Source AP (0.9311 vs. 0.8954), and Answer Relevancy (0.6195 vs. 0.6151). While BM25 (T1) scores slightly higher on lexical Answer Correctness (0.6735) due to direct keyword overlap, T4 achieves superior structural grounding and eliminates hallucinations. Ablation results confirm the vital contribution of each module: omitting the cross-reranker (T5) reduces AR from 0.6195 to 0.5263 (-0.0931) and CR to 0.8133; removing subspace governance (T7) degrades Answer Relevancy to 0.5815 and Correctness to 0.6430; and removing Knowledge Graph grounding (T6) causes a dramatic plunge in Context Precision from 0.6977 to 0.5683 (-12.75%), proving its indispensable structural value.

Scenario 3 — Multi-Agent Routing, Tool Reliability, and Robustness:

Objective & RQ Alignment: Scenario 3 directly addresses **RQ1** by evaluating multi-agent coordination, tool execution reliability, and bounded safety independently from retrieval or generation. *Experimental Setup:* Evaluated across three suites ($N = 550$ runs): (1) **Supervisor Routing (Panel A)**: 100 single-

domain benchmark queries ($N = 300$ for LLM Supervisor vs. $N = 100$ for Rule-Based Router); (2) **Specialist Tool Reliability (Panel B)**: 60 production cases ($N = 180$ across 3 reps) covering 11 production tools and no-tool gating; and (3) **Adversarial Robustness (Panel C)**: 20 failure stress tests ($N = 60$ across 3 reps) challenging boundary invariants.

Table 6. Scenario 3: Multi-agent routing, specialist tool execution reliability, and robustness under a maximum of one tool call per decision ($\text{max_tool_calls} = 1$).

Panel A: Supervisor Routing and Specialist Selection ($N = 100$ queries; Rule: 1 rep / LLM: 3 reps, $N = 300$)							
Router Variant	Runs (N)	Agent Acc.	Macro-F1	Intent Acc.	Bounded p50 (ms)	p95 (ms)	
Rule-Based Router	100	85.0%	69.2%	69.0%	100.0%	0.0	0.1
LLM Supervisor (CTU-CHAT)	300	97.0%	97.8%	95.0%	100.0%	795.5	981.7
Panel B: Specialist Tool Execution Reliability ($N = 15$ runs per function, 3 reps)							
Target Function	Execution Type	Selection	Arg. EM	Result Acc.	E2E Pass	Bounded	Graph hit
tra.cuu.nganh	LLM Tool Call	100.0%	100.0%	100.0%	100.0%	100.0%	–
so.sanh.nganh	LLM Tool Call	100.0%	100.0%	100.0%	100.0%	100.0%	–
tim.nganh	LLM Tool Call	100.0%	100.0%	100.0%	100.0%	100.0%	–
xem.chuoi.tien.quyet	LLM Tool Call	100.0%	100.0%	100.0%	100.0%	100.0%	–
mon.chung.giua.nganh	LLM Tool Call	93.3%	93.3%	100.0%	93.3%	100.0%	–
tim.nganh.co.mon	LLM Tool Call	100.0%	100.0%	100.0%	100.0%	100.0%	–
tra.cuu.hoc.phi.graph	LLM Tool Call	80.0%	80.0%	100.0%	80.0%	100.0%	80.0%
tra.cuu.co.so.mien.giam.graph	LLM Tool Call	80.0%	80.0%	73.3%	73.3%	93.3%	78.6%
tra.cuu.quy.dinh.hoc.phi	LLM Tool Call	100.0%	100.0%	100.0%	100.0%	100.0%	40.0%
tin.h.toan.hoc.phi	LLM Tool Call	100.0%	80.0%	80.0%	80.0%	100.0%	–
tin.h.tien.hoc.bong	LLM Tool Call	100.0%	100.0%	100.0%	100.0%	100.0%	–
no.tool	No Tool Call	100.0%	100.0%	100.0%	100.0%	100.0%	–
Panel C: Robustness and Bounded Failure Handling ($N = 60$ runs under malformed/adversarial inputs)							
Adversarial Gate	Runs (N)	Agent Acc.	Tool Gate Safe	Result	E2E Pass	Bounded	Notes
Failure Gate Robustness	60	100.0%	100.0%	98.3%	98.3%	100.0%	–

Note: `tuition.lookup` evaluates deterministic catalog lookup. Scholarship and fee reduction evaluate typed LLM tool selection.

Results & Interpretation: (1) *Supervisor vs. Heuristic Routing:* The LLM Supervisor attains **97.0% Agent Accuracy**, **97.8% Macro-F1**, and **95.0% Intent Accuracy**, outperforming the rule router (85.0%, 69.2%, and 69.0%) with sub-second p95 latency (981.7ms, p50 795.5ms). (2) *Specialist Tool Execution:* Five academic tools, tuition policy lookup, scholarship calculator, and no-tool gating achieved **100.0%** end-to-end pass. (3) *Robustness & Boundedness:* Under adversarial stress testing (Panel C), CTU-CHAT achieved **100.0% Tool/No-tool decision accuracy**, **98.3% Safe Result behavior**, and **100.0% bounded completion** within operational limits ($\text{max_tool_calls} = 1$, $\text{timeout} = 60$ s).

4.4 Discussion

Answering RQ1 (Domain-Specialized Multi-Agent Architecture): Centralizing routing under a structured supervisor with asymmetric privileges and bounds ($\text{max_tool_calls} = 1$, $\text{timeout} = 60$ s) produced bounded isolated decisions in the Scenario 3 runs: the LLM Supervisor achieves 97.0% agent accuracy and 97.8% Macro-F1 across 300 evaluations, outperforming rule-based routing (85.0% and 69.2%). Production-tool evaluations reached 100.0% end-to-end pass

on 8 of 12 functions, while adversarial testing demonstrated 100.0% tool-gating accuracy, 98.3% safe-result behavior, and 100.0% bounded completion.

Answering RQ2 (Heterogeneous Knowledge Allocation & The Graph Ablation): Decomposing counseling knowledge across structured relational curricula (Neo4j) and unstructured regulatory decrees (hybrid RAG) is strongly substantiated by our modular ablations. A key empirical question concerns the *without-graph ablation* (T6): *Why did T6 achieve competitive context scores on certain single-entity queries during development?* Our qualitative error analysis reveals that for **single-entity factual lookups** (e.g., standard tuition for a general major), unstructured text paragraphs alone provide complete answers; injecting structured graph nodes introduces slight context overhead and token dispersion, marginally diluting automated precision scores. However, for **multi-hop relational queries** (e.g., prerequisite academic chains, cohort-specific tuition progression, and credit-reduction criteria), the Knowledge Graph is indispensable. This is conclusively proven by the blind Held-Out Test Set: when evaluated against complex unseen queries, omitting Neo4j (T6) causes Context Precision to **plummet from 0.6977 to 0.5683 (a severe −12.75% absolute drop, −18.3% relative)**, while reducing Factual Exact Match from 62.03% to 57.11% (−4.92%) and Context Recall from 0.8933 to 0.8600. Thus, Knowledge Graph grounding provides essential structural anchors that prevent factual hallucination in complex reasoning paths.

Answering RQ3 (Representation-Aware Evidence Routing): Across cumulative retrieval configurations, top-1 retrieval rises from 0.9000 (E1) to a record **0.9600 Hit@1** (48/50 queries correct at rank 1) and **0.9700 MRR@10** for E5 on the Held-Out Test Set. In end-to-end ablation, T4’s governed Graph injection and reranking achieve the highest Factual Exact Match (62.03%), Context Precision (0.6977), and Source AP (0.9311), whereas removing governance routing (T7) degrades Answer Relevancy to 0.5815 and Correctness to 0.6430. While neural-reranked configurations exhibit higher measured latency (10.99 ms for E1 vs. 18,144.98 ms for E5), unranked hybrid fusion (E3, 43.62 ms) represents a practical low-latency alternative when computational budgets are constrained.

4.5 Threats to Validity and Limitations

While derived from Can Tho University, the administrative taxonomy generalizes across Vietnamese higher education; multi-institutional validation remains planned. In addition, cross-attention reranking latency motivates future model distillation. Curricula graphs currently rely on offline schemas, and automated judges penalize out-of-scope abstentions, underestimating practical correctness when refusing ungrounded generation. Finally, while Scenario 3 validates bounded execution and tool reliability at the isolated supervisor and specialist decision level, end-to-end fault injection across external infrastructure services (e.g., Neo4j connection drops or Qdrant timeouts) represents an orthogonal system-resilience evaluation reserved for future work.

5 Conclusion and Future Work

This paper addressed knowledge heterogeneity in university counseling, where uniform RAG pipelines and uncoordinated agent meshes can be brittle across relational curricula, structured tuition matrices, and narrative decrees. CTU-CHAT introduces a centralized supervisor-routed multi-agent architecture granting four domain specialists isolated subspaces, dedicated retrieval paths, and asymmetric tool privileges. Relational curricula are allocated to Neo4j and structured tuition to graph catalogs with deterministic engines, reserving hybrid RAG (BM25, dense embeddings, RRF, cross-encoder reranking) for narrative decrees. Specialists execute without peer-to-peer delegation, with at most one tool call per evaluated decision (`max_tool_calls` = 1).

Empirical evaluation provides evidence of 97.0% supervisor routing accuracy and 100.0% bounded completion in isolated Scenario 3 decisions. The production-tool suite achieved 100.0% end-to-end pass for academic lookup, tuition-policy, scholarship, and no-tool gates; graph tuition lookup reached 80.0%, while graph exemption-basis lookup reached 73.3% with a 78.6% Graph-hit rate. Component stacking achieves **0.9000 Hit@1**, **0.9600 Hit@3**, **0.8656 Recall@5**, and **0.9327 MRR**. With a common Top-7 context window, the full system T4 reaches the highest Answer Correctness (0.647), although the graph ablation T6 is slightly higher on context recall and precision; removing governance (T7) lowers AC to 0.482. Future work will explore cross-institutional transferability, latency distillation, and advisory trials.

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